

Fitting a curve to the dam — Results

NM-LAB-08252026 · Laboratory Activity 02 · Numerical Methods BES6-M (Engr. Mark Airol Escranda) · Submitted by Estiamba · Dataset: reservoir stage log, 15-min sampling, n = 288 readings (2026-07-21 00:00 to 2026-07-23 23:45 PHT)

Model and fit

Sum of two logistics (signed amplitudes): $h(t) = c + a_1 / (1 + e^{-k_1(t-t_{01})}) + a_2 / (1 + e^{-k_2(t-t_{02})})$. The log rises sharply to a peak near $t=36$ h and then falls off to a second, lower plateau — not a single fill settling to one ceiling — so a plain logistic or Gompertz was ruled out; the second sigmoid (negative amplitude) captures the drawdown/release after the peak. $p_0 = [14.18, 7.0, 0.5, 32.0, -1.6, 0.15, 42.0]$, read off the raw plot (baseline, rise/drawdown magnitudes, approximate inflection times). Fit via `scipy.optimize.curve_fit(method="lm")`; covariance diagonal finite and positive throughout.

Param	Unit	Value	Std. error	t stat	p-value
c	m	14.3493	0.0097	1483.16	<0.001
a1	m	8.9159	0.7532	11.84	<0.001
k1	1/h	0.5117	0.0213	24.05	<0.001
t01	h	31.5198	0.0906	348.02	<0.001
a2	m	-3.7211	0.7595	-4.90	1.62e-06
k2	1/h	0.2831	0.0138	20.58	<0.001
t02	h	37.5464	1.0576	35.50	<0.001

SSE	1.8017 m ²	R ²	0.9991	s (n-p=281)	0.0801 m
SST	2034.078 m ²	n, p	288, 7	max resid	0.221 m (30.25 h)

Reading the residuals

Residuals are centred close to zero overall (mean +0.0000 m) — no net drift — but they are not unstructured. Four runs of same-sign residuals, each 47–54 points (about 12–14 hours) long, together span almost the whole record: negative $t=0$ –11.5 h, positive $t=12.2$ –24.8 h, positive $t=43.2$ –56.5 h, negative $t=59.2$ –71.8 h. Runs this long indicate a real shape limitation, not sensor noise: the 'flat' stretches before and after the pulse are not perfectly flat in the raw log, and the constant baseline c cannot bend to follow that slow creep, so the fit rides consistently above or below it for many hours at a stretch. Spread is somewhat wider during the event (0.119 m) than in the quieter pre-event (0.086 m) and post-event (0.038 m) stretches, as expected where the curve is steepest. The largest single residual, 0.221 m at $t=30.25$ h (22× the logger's 1 cm resolution), sits right at the peak. **Verdict: R² = 0.9991 is high but sits on top of these long same-sign runs, not on unstructured scatter — an honest partial success: the two-logistic shape captures the fill/release event well but is too rigid a baseline to track the slow creep outside it.**

Derivative (Tab 1) and integration (Tab 3)

Max dh/dt (finite difference) = 0.960 m/h at $t = 29.25$ h (2026-07-22 05:15 PHT). The raw second derivative does not usefully localise the event: its spread during the fill/release window (std 0.377 m/h²) is essentially the same as on the flat parts of the log (std 0.361 m/h²) — differentiating twice turns the logger's 1 cm rounding into noise of comparable size everywhere, which is exactly why a smooth model was fit and differentiated instead of trusting the raw second difference.

Area under the fitted curve, `scipy.integrate.quad`, $t=[0, 71.75]$ h: **1260.9708 m·h** (abs. error estimate $\pm 5.10e-07$ m·h). Cross-check, `np.trapezoid` on the raw points: 1260.9963 m·h. Gap = -0.0254 m·h (-0.002%), concentrated where the straight-line trapezoid segments approximate the sharp fill/release flank poorly compared with the smooth fitted curve. The area is in meter-hours — the time-integral of stage, a measure of how high and how long the reservoir stood over the record. **It is not a volume:** converting it to cubic metres would require the reservoir's stage-to-area or stage-to-volume curve, which this stage log does not contain.